

The Floater — engineering report

$f = 0.80$ solar dynastat. As-built configuration, sealed and cargo variants, derivation chain, bill of materials, open items.

2026-08-03
site A: 47.6°N, 500 m
d195 solstice / d80 equinox

1. Summary

What it is. A helium-filled centre hull carrying 80% of the aircraft's weight, flanked by twin solar wings that fly the remaining 20% and a battery that carries the night. One 100 W-class electric thruster. There is no fuel and no engine noise beyond a 12-inch prop turning at cruise power; on a clear day the aircraft harvests more than it spends and the pack ends the day where it started. That is the whole design intent: **perpetual on sunlight, near-silent, and buildable from parts you can order today.**

Headline. The as-built aircraft — the one BOM_v2.csv actually buys — **closes its 24 h energy loop at day 195 on a 98.0 Wh pack, at 2272.4 g all-up, on a 2991.4 × 997.1 mm (117.8 × 39.3 in) LLDPE hull of 1.557 m³.** The parts cost **\$518.11** plus gas (**\$738.11** all-in on party-tank helium, **\$581.10** on hydrogen with a budget electrolyzer). The pack actually on the BOM (pack A, 80.6 Wh) **misses by 11.6 Wh**; two more Molice P28A cells, **\$19.90**, clear it.

What is certified and what is not. Every number in this report is an *ideal-chain* or *piecewise-path* result: the flight-physics engine's real-drive chain (BEMT propulsion + buoyant trim at real tolerances) **is red**. Five defects are open (STATUS §5): the BEMT certification envelope refuses the lightly-loaded prop at the top of the trim bracket; buoyant trim cannot converge because the solver's tolerance is set against *gross* weight while the wing only controls the (1-f) residual; the aero polar refuses at the 4.4–4.5 m/s buoyant trim speed; pack thermal is never stepped; and four verification anchors were never evaluated. **None of those defects is an energy, chemistry, materials or mission finding** — flipping one line back to the ideal drive reproduces the pre-upgrade gate bit-identically, which isolates the whole regression to the drive and its certification envelopes. So: **the mass ledger, the hull sizing, the harvest and the 24 h closure are the numbers to trust; the propulsion and trim results await a re-certification on the real chain.** Engine work is ongoing.

What to build. Config 2, the as-built LLDPE aircraft, with a 10-cell (100.8 Wh) pack. It is the only configuration in the study that is both fully sourced *and* closes. Everything else — sealing for six months, tilt-thrust, cargo — is an upgrade to this same airframe, and each one has a live blocker: the 6-month barrier film has no small-lot vendor, the 6-month seal margin is –1.8 Pa at day 180, and a 10 lb loiterer is a different aircraft ($f \approx 0.95$, 8.2 m³), not a stretch of this one.

The one part nobody has designed is the vent / ballonet. It gates any sealed build and it is not in any result file. Until it exists, fly the hull unsealed, on short supervised line-of-sight windows, and never leave it sealed on the ground in daylight.

2. Top designs

Numbers are the corrected as-built run (V2_10_asbuilt.out / _result.json) and STATUS §2. **The earlier V2_02_asbuilt run is contaminated and its output is retracted** — it silently inherited sealed-laminate film defaults (0.797 kg of film on a 1.99 m³ hull, ~45 g/m²) while BOM_v2.csv actually buys 38 µm LLDPE at ~35 g/m². Its headline — 4.016 kg all-up, 3.62 × 1.21 m hull, 234.6 Wh minimum pack — is wrong by construction. Do not use it.

	A — AS-BUILT build this first	B — SEALED 6-MONTH	C — 1 LB DELIVERY	D — 10 LB CARGO
Hull	2991.4 × 997.1 mm (117.8 × 39.3 in), 1.557 m ³	3568.2 × 1189.4 mm (140.5 × 46.8 in), 2.643 m ³	unchanged as-built hull	5214.6 × 1738.2 mm, 8.249 m ³ (loiter, f = 0.95)
Film	38 µm LLDPE	45 g/m ² PET12/PE30 laminate	38 µm LLDPE	45 g/m ² laminate
Wing	2.5 m / 0.52 m ² / AR 12 / 21 cells	3.5 m / 1.019 m ² / 41 cells	2.5 m / 21 cells	3.5 m / 41 cells
All-up	2272.4 g	3857 g	2272 g + 454 g package	10.575 kg incl. 4.536 kg package
Pack, d195	98.0 Wh min (pack A 80.6 Wh misses by 11.6 Wh)	122.4 Wh	— daylight only	165.2 Wh at f = 0.95
Pack, d80	180.3 Wh	203.5 Wh	—	not run
24 h loop	closes at 98 Wh	closes BOTH days	does not close with the package aboard	no pack closes at f = 0.80
Buildable today	yes — every part sourced	no — film is a spec, not a part	yes	no — different airframe
Cost	\$518.11 core + gas (+\$19.90 for the closing pack)	+\$91 solar, film quote-only	as A	not costed

A — AS-BUILT FLOATER (build this first)

The aircraft BOM_v2.csv buys, at real measured part masses, with the hull re-sized to hold f = 0.80. Hull **2991.4 × 997.1 mm = 117.8 × 39.3 in** (9 ft 9.8 in long), **1.557 m³**, 7.68 m² of skin, 38 µm LLDPE. Wings 2.5 m span / 0.52 m² / AR 12, twin, flanking the hull. All-up **2272.4 g**, trim 4.00 m/s, total load 9.18 W (prop 6.43 + payload 2.76); harvest 615.9 Wh/day at d195, 391.6 Wh/day at d80.

Where the +274 g came from — the honest core of this report. Against the v1 certified ledger: the real DIY pack lands at **197 Wh/kg, not the certified 250 (+92 g)**; the real thruster group — motor + ESC + prop — is **120 g, not the certified 55.2 g (+65 g)**; and the certified ledger carried **no MPPT at all**, where the Genasun GV-5 is **+57 g**. That is +214 g of parts. The remaining **~60 g is the hull growing to lift them** — a bigger envelope needs more film and more gas, which needs more envelope, and the fixed point converges at 2272.4 g. It is also 5 inches longer and 1.6 inches fatter than the design already drawn (+125.6 mm length, +41.9 mm diameter, +0.188 m³, +0.20 m/s trim, +1.32 W load).

Closure. Real **pack A (80.6 Wh, 8 × P28A, 410 g) does not close** — it goes through the 0.05 SOC floor with **11.6 Wh unserved** at d195 and is 42.1 Wh short at d80. The **minimum closing pack is 98.0 Wh** (0.497 kg, all-up 2385 g, hull grows to 3040.0 × 1013.3 mm = 119.7 × 39.9 in, 1.634 m³). **Pack B (121.0 Wh, 605 g) closes d195 outright, zero unserved, SOC floor 0.164** — all-up 2523.8 g, hull 3097.9 × 1032.6 mm = 122.0 × 40.7 in, 1.730 m³, trim 4.12 m/s, load 10.40 W — but it is still 21.3 Wh short at equinox, which needs **180.3 Wh** (2919 g, 3251.9 × 1084.0 mm). **The fix is two more P28A cells** — 2S5P, 10 cells, 100.8 Wh, ~512 g, **+\$19.90** — which clears the 98.0 Wh minimum with **2.8 Wh to spare**. It is the cheapest gap in the entire study.

One caution. Pack B dropped into the pack-A hull does *not* work: f falls to 0.735, trim rises to 4.70 m/s and the loop opens. **Size the hull for the pack you will actually fly.**

B — SEALED 6-MONTH

Six months without a refill costs **+70% all-up mass and a metre more wingspan**: 3857 g, 3.5 m / 41 cells, hull 3568.2 × 1189.4 mm (2.643 m³), 471 g of helium. It closes **both** days — 122.4 Wh at d195, 203.5 Wh at d80 — and it is the only configuration in the study that does. Three conditions attach to that result.

- **It needs the 45 g/m² film, and the wing cannot substitute.** A span sweep 2.5 → 5.0 m bottoms the pack-B shortfall at b = 3.5 m (18.6 Wh) and then it *rises* again — past 3.5 m the added night load outruns the added daylight. Changing the film spec from the assumed 70 g/m² to the sourced 45 g/m² is worth **110 Wh of pack and 1.17 kg of aircraft** by itself. The film was the constraint, not the wing.
- **That film has no small-lot vendor** (flag H2). IMPAK, Sorbent Systems, Caltex and Goodfellow all quote-only; Alibaba mills want 500–2000 m². The one purchasable substitute, CS Hyde 25.4 µm metallized PET at \$7.83/m², **has no sealant layer** — seams must be taped, not heat-sealed. This configuration's headline rests on a film you cannot currently buy in the right form.

- **The seal margin is zero** (flag H3). At $100 \text{ cm}^3/\text{m}^2/\text{day}/\text{atm}$ the hull loses 7.67% of its gas in 180 days and reaches **-1.8 Pa** superpressure on a cold night at day 180 — slack exactly at the six-month mark. **Carry +15% gas, not +10%.**

C — 1 LB DELIVERY

The unchanged as-built hull with a 0.454 kg package added; the hull is *not* re-sized, so f falls to 0.642 on the 98 Wh pack. **Lift is never the problem** — it trims and flies at **cruise 5.94 m/s** drawing 16–21 W of propulsive power against a 100 W drive, roughly 20% of installed thrust. **Dash is 12 m/s**, the fastest point inside the 100 W rating. A **16.1 km (10 mi) leg** takes 0.75 h at cruise on 16.2 Wh, or 0.37 h at dash on 38.0 Wh; a round trip at cruise is 32.4 Wh against 88.4 Wh of usable pack, so both legs fly on one charge with no sun. Daylight at d195 is 15.07 h and harvest is 615.9 Wh/day, giving 19.0 energy-limited trips against 10.0 time-limited ones — **the binding answer is 10 round trips per day, 322 km of delivery flying.**

The night loop does not close with the package aboard — 119.6 Wh short at d195 on the 98 Wh pack. Carrying a pound overnight costs about 12 W of extra cruise power, more than this airframe's night budget. **This is a daytime shuttle, not a persistent loiterer:** it must drop or land before sunset. Below 6 m/s the polar has no valid row at the required CL and the model refuses rather than extrapolating — those rows are printed as refusals.

No drop problem at 1 lb. Gross lift is 1.818 kg against 2.378 kg of remaining weight, so releasing the package leaves the aircraft 0.560 kg heavier than air. It does not rise; it simply flies more easily as the wing reverts to the 1.014 kg residual it was already carrying.

D — 10 LB CARGO

It does not loiter at $f = 0.80$. With the package aboard the hull re-sizes to $4949.1 \times 1649.7 \text{ mm}$ (7.052 m^3 , 21.02 m^2 of skin, 1312 g of helium), all-up 10.736 kg, cruise 8.46 m/s, load 122.58 W. The 24 h budget is **2942 Wh required against 600 Wh harvested** — a factor of five, not a marginal miss, and there is **no closing pack at any size**. Growing the wing to 3.5 m helps the trim a lot ($8.46 \rightarrow 6.44 \text{ m/s}$, $122.58 \rightarrow 71.68 \text{ W}$) and still does not close. f is the wrong knob: it leaves the wing flying 2.32 kg of residual on 1.02 m^2 , a wing loading the planform can only meet by flying fast, and drag goes as V^2 .

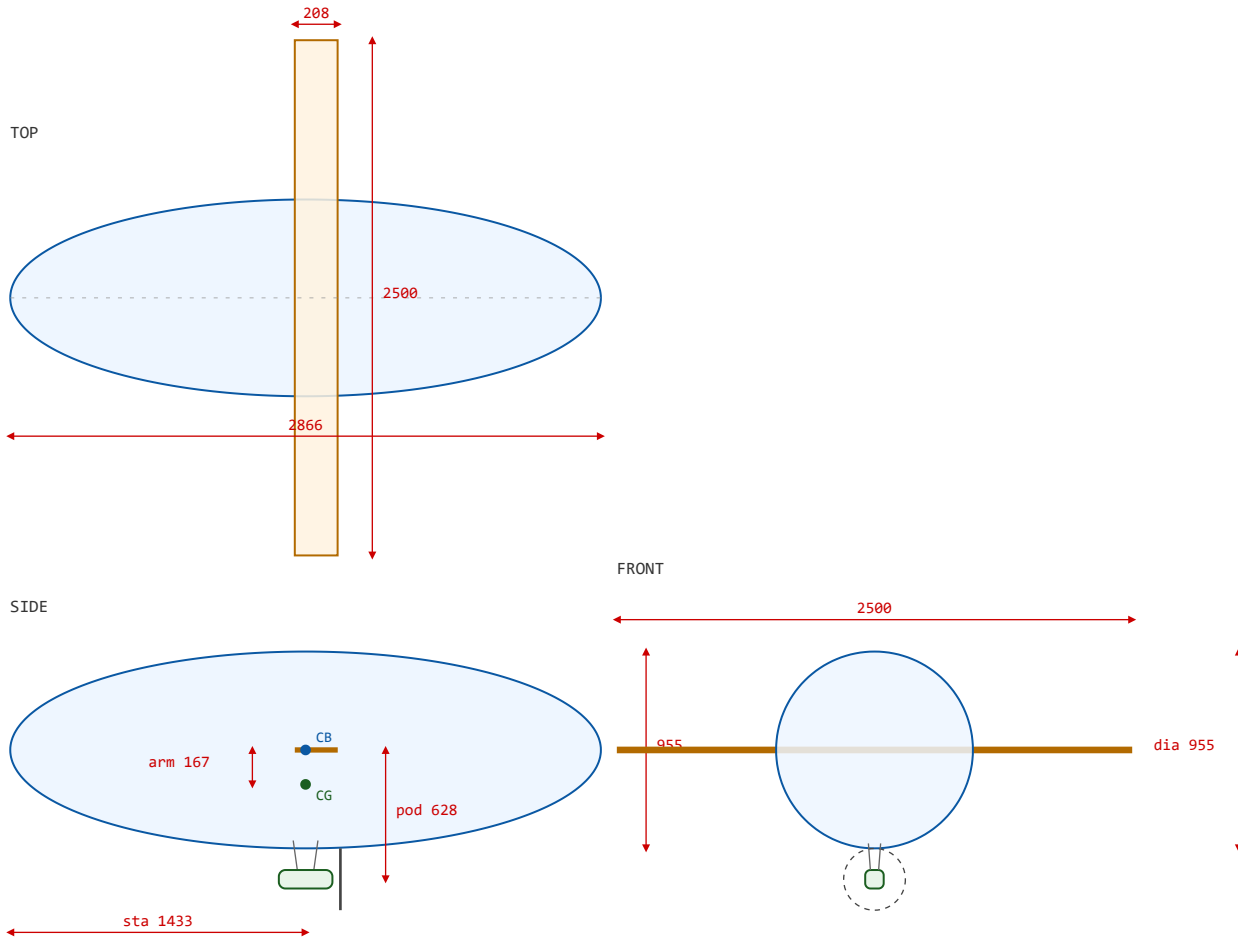
It loiters perpetually at $f \approx 0.95$ — 165.2 Wh pack, 10.575 kg all-up, 8.249 m^3 hull, cruise 2.95 m/s, load 16.26 W. Going $0.80 \rightarrow 0.95$ costs 0.6 m^3 of hull and saves 1.05 kg and 55 W. For cargo the aircraft wants to be an airship with a wing, not a wing with a balloon.

Delivery works either way: 16.1 km in 0.53 h on 64.8 Wh, round trip 129.6 Wh against 361.0 Wh usable — both legs with zero sun.

The drop problem, at 10 lb. Gross lift 8.588 kg against 6.200 kg remaining leaves **+2.389 kg (23.43 N) of net upward force the instant the package leaves**. A constant-volume hull must then climb until $\rho_{\text{air}} = 0.879 \text{ kg/m}^3$, with superpressure rising the whole way — **the envelope bursts long before equilibrium; release without mitigation is hull loss**. Ballast exchange (take on $\geq 2.389 \text{ L}$ of water at the drop point) is the only economically sane answer and the only one that scales to repeated drops. Thrust-down holding needs the 400 W drive and burns 398.1 W — 54 minutes on a full 400 Wh pack, a delaying action. Venting sheds 2.278 m^3 , **32.3% of the hull**, at \$97–161 per drop, after which any sealed claim is over. **Every drop point needs water.**

3-view

f=0.80 SOLAR DYNASTAT - 3-VIEW (dimensions mm, from design_snapshot.json)



hull 3:1 prolate spheroid 2866 x 955 mm (V 1.369 m³) - wing 2500 x 208 mm Tier-1 CST - pod slung 628 mm below axis at the CB station - prop dia 300 mm
CB above CG 167 mm (pack A). Pod drawing size is schematic [choice]; all dimensioned values from design_snapshot.json.

Note on scale. This drawing is the **v1 catalogue geometry** from design_snapshot.json — hull 2866 × 955 mm, 1.369 m³, 1998 g. The as-built aircraft in §2A is the same shape and the same wing at a larger hull scale: **2991.4 × 997.1 mm, 1.557 m³, 2272.4 g** (+125.6 mm length, +41.9 mm diameter, +0.188 m³). Wing, section, prop diameter, pod station and the CB-over-CG arrangement are unchanged; the STLs, gores and ribs already cut are still the right shape. Only hull scale and pack size move.

3. The math

The derivation chain, with our numbers substituted. Values marked *[run]* are read from a result file; *[derived here]* means the relation was evaluated in this report from run inputs.

E1 — Atmosphere (ISA troposphere + Sutherland viscosity)

$$\begin{aligned} T &= T_0 - L \cdot h & p &= p_0 \cdot (T/T_0)^{(g/(R \cdot L))} & \rho &= p / (R \cdot T) \\ \mu &= \mu_0 \cdot (T/273.15)^{1.5} \cdot (273.15 + 110.4) / (T + 110.4) \end{aligned}$$

Every aerodynamic and buoyancy number below is anchored to one atmospheric state. At the design altitude of **500 m**: $T = 288.15 - 0.0065(500) = \mathbf{284.900 \text{ K}}$, $p = \mathbf{95 \text{ 461.3 Pa}}$, $\rho = \mathbf{1.16727 \text{ kg/m}^3}$ *[run]*, and Sutherland gives $\mu = \mathbf{1.7737e-5 \text{ Pa}\cdot\text{s}}$. That

μ is not an assumption: back-solving it from the engine's own reported Re anchor (51 963.9 at 3.7961 m/s, chord 0.208 m) returns 1.77365e-5 — agreement to five digits, so the report and the engine are on the same atmosphere.

E2 — Reynolds number

$$Re = \rho * V * c / \mu$$

Sets which polar rows are valid. The wing chord is **0.208 m** (2.5 m span, 0.52 m², AR 12.02, constant chord). At the as-built trim of **4.00 m/s**: $Re = 1.16727 \times 4.00 \times 0.208 / 1.77365e-5 = 5.48e4$ [derived here]; the v1 anchor is **Re = 51 964** at 3.796 m/s [run]. Both sit well above the polar's 30 000 floor, which matters: the trim refusals reported in §5 at 4.4–4.5 m/s correspond to $Re \approx 59\,000$, so **those refusals are an angle-of-attack / model-confidence limit, not a Reynolds floor**. This is squarely low-Reynolds flight — sailplane-model territory, where profile drag is high and section data is scarce.

E3 — Buoyancy and the fraction f

$$\begin{aligned} L_{gross} &= V_{gas} * (\rho_{air} - \rho_{He}) * g, & \rho_{He} &= 0.13819 * \rho_{air} \\ f &= \rho_{air} * V_{gas} / M_{allup} & \text{residual the wing must fly} &= (1 - f) * M_{allup} * g \end{aligned}$$

f is **gross Archimedes lift over all-up weight** — it counts the displaced air, and the gas mass sits inside M_{allup} . For the as-built hull: $1.16727 \times 1.5574 = 1.8179$ kg of air displaced, against an all-up of **2.2724 kg**, giving **f = 0.800** [run, and reproduced here]. The gas at the sizing state ($\rho_{gas} = 0.16215$ kg/m³, ambient +500 Pa superpressure) masses $0.16215 \times 1.5574 = 252.5$ g, exactly the ledger line. Net static lift is therefore $1.5574 \times (1.16727 - 0.16215) = 1.5654$ kg against 2.0199 kg of non-gas aircraft — the ship rests **0.4545 kg = 4.457 N heavy**, which is precisely $(1 - 0.80) \times 2.2724 \times 9.80665$. **That 4.46 N is the entire load the wing flies**. The v1 aircraft is the same arithmetic one size down: $1.16727 \times 1.3694 = 1.5983$ kg over 1.9981 kg = f 0.800, residual 3.92 N (about 400 gf, the weigh-off target on the build sheet).

E4 — Level-flight power, and why buoyancy collapses it

$$\begin{aligned} P &= \sqrt{2 * W^3 / (\rho * S)} * C_d / C_l^{1.5} / \eta & (P \text{ scales as } W^{1.5}) \\ \text{engine's chain: } P_{prop} &= (D_{wing} + D_{env}) * V / \eta_{drive} \end{aligned}$$

The wing does not fly the aircraft's weight; it flies the **(1-f) residual**. Because power goes as $W^{1.5}$, taking the wing's load from the gross 22.29 N down to 4.457 N cuts its required power by $5^{1.5} = 11.2\times$. That single factor is why a 2.3 kg aircraft cruises on single-digit watts, and it is the whole reason the design is buoyant at all.

The price is the envelope. At the as-built d195 point [run]: **D_wing 0.33 N, D_env 0.55 N** — the hull is now **63% of total drag**. Mechanical power is $(0.33 + 0.55) \times 4.00 = 3.52$ W; the run reports **6.43 W** of electrical propulsive power, so the implied prop $\times$ motor $\times$ ESC efficiency is **0.55** [derived here]. Total load is **9.18 W = 6.43 prop + 2.76 payload**. Party helium is the same equation run backwards: residual 4.46 $\rightarrow$ 7.55 N raises prop power 6.43 $\rightarrow$ 12.51 W, a 69% penalty for no extra useful load.

E5 — Solar harvest

$$P_{peak} = G * A_{wing} * \text{packing} * \eta_{cell} \qquad A_{active} = 0.52 * 0.6923 = 0.36 \text{ m}^2$$

21 SunPower C60 cells on the 0.52 m² wing at the v1 packing fraction of 0.6923 give **0.36 m² active** at **$\eta = 0.227$** , so at 1000 W/m² STC the array peaks at **81.7 W** [derived here]. Integrated over the d195 solar day at 47.6°N / 500 m the run returns **615.9 Wh/day** [run] — a 40.9 W daylight average across 15.07 h of sun, half of peak. At equinox (d80) the same array returns **391.6 Wh/day**, and that 224 Wh/day collapse is why every configuration needs a substantially bigger pack for d80 than for d195. **The array was never the v1 constraint**: that aircraft **spilled 423.7 Wh** of the 615.9 it harvested (68.8%), because the pack was full and there was nowhere to put it. It becomes a constraint only once the sealed film and the real part masses push the ship into a harvest-limited regime — which is exactly why the sealed configuration needs 41 cells and the as-built one does not.

E6 — Limit-cycle closure

$$\begin{aligned} \text{closes iff } SOC(t_{end}) &\geq SOC(t_{start}) \quad \text{AND} \quad \text{min SOC over the window} > SOC_{MIN} \\ SOC_{MIN} &= 0.05, \quad \eta_{charge} = 0.95, \quad \eta_{discharge} = 0.95, \quad \text{scored on the SECOND 24 h window} \end{aligned}$$

Why the first window lies. The first 24 h begins on a pack the initial condition handed over full. That stored energy was never earned inside the window, so an aircraft running a genuine daily deficit can still serve its night out of the gift and finish looking healthy. The second window has to buy its own night out of the day it just flew — it is the first window that carries no subsidy, and it is the one scored.

Our numbers [run]: as-built pack A at d195 ends with **11.6 Wh unserved** and **min_soc reading exactly 0.050**. That is not a margin — the solver clamps at the floor, so "0.050 with unserved > 0" means the aircraft went *through* it. The smallest pack

that satisfies both tests is **98.0 Wh**; pack B at **121.0 Wh** satisfies them with **min_soc 0.164**, the only as-built point in the study with real daylight above the floor. Equinox (d80) needs **180.3 Wh**.

E7 — Party-helium check (negative result)

```
rho_mix/rho_air = 0.8*(0.13819) + 0.2*(1.0) = 0.31055
net lift ratio = (1 - 0.31055) / (1 - 0.13819) = 0.68945 / 0.86181 = 0.8000 (exactly x0.80)
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Balloon Time's own wording is "not less than 80 percent helium"; the balance is air. Run in the *same physical hull* sized for 99.9% [run]: gas mass **253 → 568 g**, all-up **2.272 → 2.587 kg**, and therefore **f = 1.8179 / 2.587 = 0.703** — not the 0.64 the brief predicted. The lift per cubic metre *does* fall by exactly $\times 0.8000$, but **f does not scale with lift**, because $f = \rho_{\text{air}} V / M_{\text{allup}}$ and the blend makes M_{allup} larger. The consequence is worse than the mechanism sounds: the residual the wing must fly rises **4.46 → 7.55 N (+69%)**, trim **4.00 → 5.17 m/s**, total load **9.18 → 15.52 W**. **It does not fly, and no pack in this airframe closes it. Welding-grade 99.9% only.**

4. Bill of materials

From BOM_v2.csv (as-built, pack A). Prices are indicative vendor-verified estimates in USD. Masses are real measured or vendor-stated part masses, not ledger allowances.

item	part	qty	mass g	\$	honesty
PROPULSION — 120 g as a group					
motor	SunnySky X2212 KV980 (V3)	1	56	15.30	certified thruster ledger is 55.2 g for motor+ESC+prop <i>total</i> ; this motor alone is 56 g
propeller	APC 12×8E Thin Electric (LP12080E)	1	26	4.89	12 in = the certified 0.30 m dia; fixed, not folding — stopped-prop drag is not in the drag ledger
ESC	Hobbywing Skywalker 30A (2-3S)	1	38	17.32	2–3S variant only (the V2 is 3–4S); $\sim 2\times$ current margin at 100 W on 2S
ENERGY					
battery pack A	Molicel INR18650-P28A $\times 8$, 2S4P	8	410	79.60	80.6 Wh. DIY pack lands 197 Wh/kg , not the certified 250 — +92 g
<i>pack B (alt)</i>	<i>P28A $\times 12$, 2S6P</i>	12	605	119.40	121.0 Wh — 3.6% under the 125.5 Wh v1 certified figure, carried honestly (H10)
solar cells	SunPower C60/E60 5×5 flexible mono	21	234	94.50	234 g includes the 0.45 kg/m ² laminate on full wing area; bare cells are 147 g. Buy 2–3 spares — C60s crack
MPPT	Genasun GV-5-Li (2S li-ion variant)	1	57	100.00	0 g in the certified ledger. Buck topology needs $V_{\text{panel}} > V_{\text{batt}} \rightarrow$ works only with the 2S pack; confirm the 2S variant before ordering
AVIONICS — 82 g, inside the 100 g budget					
flight controller	Matek F405-WING	1	25	60.00	iNav / ArduPilot, 6–30 V input so 2S is fine
GPS	Beitian BN-220 (GPS+GLONASS)	1	5	18.00	—
servos	TowerPro MG90S metal-gear	2	27	12.00	the "9 g" class actually weighs 13 g each with lead
STRUCTURE + ENVELOPE					
spar + boom	3K carbon tube 10×8×1000 mm	3	120	25.50	2 tubes + joiner = the 2.5 m spar at 30% chord; 3rd is pod-boom stock. Inside the 738 g wing budget
hull film	1.5 mil (38 μm) LLDPE sheeting, 12 ft $\times$ 100 ft	1	271	66.00	need $\sim 10\text{ m}^2$ of a $\sim 111\text{ m}^2$ roll. Commodity "painter's" rolls are LDPE, not LLDPE — buy packaging-grade tubing if provenance matters
seam tape + valve	3M 8898 / Tesa PSA + 10 mm barb	1	0	25.00	mass already inside the $\times 1.10$ film factor
CORE PARTS (pack A, before gas)				518.11	
LIFT GAS — choose one					
helium, party tanks	Balloon Time Jumbo 14.9 cu ft	4	222	220.00	99.9% variant ONLY — the "not less than 80%" blend does not fly (E7). $\sim 0.38\text{ m}^3$ usable per tank

helium, cylinder	125 cu ft owned cylinder + Grade 4.5+ fill	1	222	510.00	99.995% pure; 3.54 m ³ = 2.6 hull fills. Cheaper per fill after the first
hydrogen, budget	generic 1 L/min SPE electrolyzer	1	114	62.99	gas mass 114 g vs He 222 g <i>and</i> +8% lift/m ³ . Flammable 4–75% in air; fill outdoors, bond and ground. A lift-gas swap needs its own weigh-off
hydrogen, lab	MSE PRO 1 L/min PEM	1	114	3500.00	price on quote; same gas as the \$62.99 unit, laboratory build quality

Cost summary — exactly as certified

build	total	= core + gas
core parts (pack A, no gas)	\$518.11	—
hydrogen + budget electrolyzer	\$581.10	518.11 + 62.99
helium, party tanks	\$738.11	518.11 + 4 × 55.00
helium, welding cylinder	\$1,028.11	518.11 + 510.00 (cylinder + first fill)
hydrogen + lab PEM electrolyzer	\$4,018.11	518.11 + 3,500.00
REQUIRED UPGRADE: +2 Molicel P28A → 2S5P, 100.8 Wh, ~512 g	+\$19.90	the only change that turns a non-closing aircraft into a closing one
pack B instead of pack A (121.0 Wh, 605 g)	+\$39.80	4 extra cells

Running cost per fill

Hydrogen: a 1.4 m³ fill at 1 L/min takes **~23 hours** and about **7 kWh** — roughly **\$1 of electricity**, and the gas is essentially free thereafter. That is the argument for the electrolyzer: it converts a recurring gas bill into a one-time \$63 (or \$3 500) capital line. **Helium:** about **\$150–250 per welding-cylinder refill** (~\$150 used in the totals), or a fresh set of party tanks. The comparison matters because **the LLDPE hull is not sealed — it refills every few weeks**, so gas is the dominant operating cost of the as-built aircraft and the entire economic case for the sealed configuration.

5. Honesty and open items

The vent / ballonnet is not designed. This is the single component gating a sealed build.

The thermal ledger prices the need — sealed peak-day superheat +25.6 K drives **+9.1 kPa** of superpressure on this hull class, about **81 MPa of hoop stress in 38 μm film, far beyond LLDPE**; at night the same ledger goes to **-1.5 kPa** and the hull goes slack and loses shape. No vent sizing, ballonnet volume, or slack-hull flight behaviour exists in any result file. Until that work lands: **fly unsealed, short supervised line-of-sight windows in mild sun, and never leave the envelope sealed on the ground in daylight**. Modelling the sealed config's +10% gas reserve as extra moles in a fixed volume also makes it a genuine superpressure balloon (10.1 kPa at fill, 19.6 kPa on a hot afternoon — 20–40 $\times$ the 500 Pa v1 hull), and the 0.011 kg/m² tendon line is an *allowance, not a structural design*.

Engine defects — the real-drive gate is RED

All five are open and being worked. Note what they are *not*: flipping one line (build_solar_cruise's chain default back to "ideal") reproduces the pre-upgrade gate bit-identically, which isolates the entire regression to the real drive and its certification envelopes. **Chemistry, electrical, materials and mission are clean.**

- **B1 — BEMT certification envelope too tight.** At $V = 12.00$ m/s the lightly-loaded prop hits high advance ratio and **98% of the thrust integrand falls outside the polar's gates**; the swirl tolerance is $1\text{e-}8$ against a $2.17\text{e-}3$ residual, five orders tighter than the physics warrants. Refuses every real-drive flight.
- **B2 — buoyant trim cannot converge, and it blocks the flagship.** Vertical residual $5.50\text{e-}4$ N against a $1.476\text{e-}7$ N tolerance — **$3.7\text{e-}5$ relative, physically meaningless and numerically fatal**. An independent sweep of $f = 0.2/0.4/0.6/0.8$ on a design that is fine at $f = 0$ refuses all four, with the residual scaling with f . Root cause: the tolerance is $\sim 1\text{e-}8$ relative to *gross* weight while the wing only controls the $(1-f)$ residual. **The certified Floater cannot currently be evaluated by the main engine.**
- **B2b — uncertified aero at buoyant trim speed** ($\sim 4.4\text{--}4.5$ m/s) on both chains. $Re \approx 59\,000$ is above the 30 000 floor, so the refusal is alpha-span or model confidence, not Reynolds. Needs precise diagnosis before anything is changed.
- **B3 — pack thermal is dead.** Over 72 h $\text{pack_T_min} == \text{pack_T_max} == 288.15$ K exactly and $\text{heater_Wh} == 0.0$ exactly — 288.15 K is the initialiser, and ambient at 500 m is 284.9 K. The thermal ODE is never stepped from the mission, which silently deletes the cold-night physics the upgrade existed for.
- **B4 — four verification anchors never evaluated** (harness import-name errors, not code defects): cold usable/ R_{int} ratios, C60 STC worst-bin MPP, He permeation through 38 μm LLDPE, and mission ledger reconciliation. Verification is incomplete.

What did survive realism: the Tier-1 wing flew a full 72 h on the real chain with an ideal drive — min SOC 0.0500, end SOC 0.4765 — and realism cost only 2.4 points of usable capacity versus the ideal chain. Hidden losses the flat model omitted, over 72 h: MPPT tracking 42.02 Wh, MPPT converter 29.57 Wh, harness I²R 6.60 Wh, night parasitic 0.19 Wh; prop/motor/ESC rose 7.6%. Battery aging to 80% capacity on this duty cycle is **2865 days (7.8 years)** — the pack is not the life-limiter.

Mass and sourcing gaps carried openly

- **DIY pack density is 197–200 Wh/kg, not the catalogue 250.** Costs +92 g on pack A and +103 g on pack B. Samsung INR21700-50E would give 232 Wh/kg at pack level — the cheapest mass lever in the BOM — but the as-built certification is on P28A.
- **The real thruster group is 120 g, not the 55.2 g in the certified ledger** (+64.8 g). The 56 g motor alone exceeds the ledger's whole allowance for motor + ESC + prop.
- **MPPT was 0 g in the ledger.** The Genasun GV-5 is +57 g — an entire component class that the certified mass budget did not contain.
- **The 45 g/m² heat-sealable laminate has no small-lot vendor** (H2). Quote-only at IMPAK, Sorbent Systems, Caltex and Goodfellow; 500–2000 m² MOQ from Alibaba mills. The one purchasable film (CS Hyde met-PET, \$7.83/m²) has no sealant layer. **The sealed configuration's headline rests on a film that is a specification, not a part.**
- **The 6-month seal has zero margin** (H3): -1.8 Pa superpressure on a cold night at day 180. Carry **+15%** gas, not +10%.
- **Welding-grade helium only.** The 80%-purity party blend costs 69% more power and does not fly (E7). Confirm the label before buying, or use the cylinder route.
- **Airspeed sensing is unsolved on this airframe** (H6). The aircraft cruises at 3.7–6.3 m/s where dynamic pressure is 10–24 Pa; the budget MS4525DO is ± 6895 Pa full scale and even the recommended DLVR-L10D at ± 2500 Pa is marginal. This is not a \$50 line item.

- **The regen ESC is billed at +0 g and that is wrong** (H5). The sourced VESC-class controller is 80 g and replaces the 38 g Skywalker — a real +42 g and +\$38.68, not in the sealed ledger.
- **The prop is held at 0.30 m in every configuration** (H11), including the 10 lb cargo case. A real 11 kg cargo aircraft wants a bigger disc, so **the cargo numbers here are conservative on that axis**. Wing structural mass (H13) is a single-anchor regression calibrated to AtlantikSolar AS-2, range-checked but not a structural analysis. Superheat (H9) comes from a sphere thermal model applied to an ellipsoid hull.
- **Iridium is a subscription, not a part** (H4): \$17/mo rental plus \$72–144/mo for hourly position reports, or ~\$305/mo at 15-minute cadence.

Discrepancies between the source documents — reported, not reconciled

1. **Sealed 70 g/m² headline.** STATUS §2.2 states the spec-film sealed config "closes d195 at 134.4 Wh (all-up 2.916 kg, hull 3.25×1.08 m)". V2_CONFIG §3 gives that same configuration ($b = 2.5$ m, 70 g/m²) as **250.4 Wh / 4.165 kg / 3.66 × 1.22 m**. The 2.919 kg / 3251.9 × 1084.0 mm figures in STATUS match the *as-built d80 minimum-pack* run instead. STATUS's sealed line appears to have crossed two runs; V2_CONFIG's values are used in this report.
2. **Equinox load for the same sealed point:** STATUS §2.2 says 15.98 W × 24 h = 383 Wh; V2_CONFIG §2/§3 say 16.05 W and 385 Wh.
3. **Spill fraction:** V2_CONFIG §2 says v1 "spilled ~60%" of its harvest; design_snapshot.json gives 423.7 of 615.9 Wh = **68.8%**.
4. **Solar active area:** 21 C60 cells at 125–127 mm square total 0.328–0.339 m², but the ledger's 0.36 m² "active" comes from packing (0.6923) × wing area — about 6% apart.
5. **Pack A mass:** 409.1 g in the V2_CONFIG ledger, 410 g in BOM_v2.csv (hence +90.9 g vs the +92 g quoted in STATUS §7).
6. **BOM_v2.csv row 15 still carries the retracted party-helium figure** — it says f "becomes ~0.64". The certified answer is **0.703** (H1 / E7).
7. **Two different helium-cylinder routes are priced.** BOM_v2.csv uses a 125 cu ft cylinder at \$510 (the number inside the \$1,028.11 total); BOM_v2_deltas.csv found a 219 cu ft cylinder + regulator at **\$207**. The cheaper route is not reflected in any total. Party tanks are likewise \$55 each in BOM_v2.csv and \$75.99 each in the deltas file.
8. **Two different "shortfall" numbers for pack A** are both correct and easily confused: **11.6 Wh** is unserved energy at d195; **17.4 Wh** is the gap between the 80.6 Wh pack and the 98.0 Wh minimum.

Sources, all on disk: STATUS.md (state of record, 2026-08-03), V2_CONFIG.md (three configurations certified, 41 KB), BOM_v2.csv, BOM_v2_deltas.csv, design_snapshot.json, BUILD_SHEET.md, three_view.svg. Nothing in this report was re-simulated; the only computed values are those marked *[derived here]* in §3, each of which reproduces a run-reported anchor. Site A = 47.6°N, 500 m, day 195 (solstice side) and day 80 (equinox).